

Sarthak Talwadkar

Boston, MA | (857) 351 8032 | sarthak.talwadkar@outlook.com | <http://linkedin.com/in/sarthak-talwadkar> | [Sarthak](#)

Skills

Programming:	Python, Java, C++, ROS, Matlab, Javascript, React
Databases:	SQL, PostgreSQL, MongoDB, VectorDB
Tools & Platforms:	Computer Vision, SLAM, Pytorch, Tensorflow, Solidworks, REST, Git, Docker, GCP, Spring MVC
Libraries:	OpenCV, Keras, Pandas, Numpy, scikit-learn
Infrastructure:	Linux Administration, Active Directory
Certifications:	Google Cloud Engineer(GCP)

Experience

Tata Consultancy Services Ltd <i>System Engineer</i>	December 2021 – July 2024 Mumbai, IN
- Designed and implemented a scalable microservice architecture using Spring Boot, incorporating robust authentication mechanisms such as OAuth2 to enhance security, performance and reliability of web applications, delivering a 40% reduction in application processing time and enhancing system scalability for 100,000+ concurrent users - Integrated enterprise-grade cryptographic solutions including AES-256 encryption and secure key management, achieving full compliance with ISO 27001 and PCI-DSS standards while eliminating security breaches - Revamped and implemented secure REST APIs using Spring Boot for seamless third-party integrations, driving an 80% increase in monthly revenue and expanding market reach by 3 times - Devised a user credibility prediction model using machine learning algorithms and 10TB+ historical data, incorporating data pre-processing, incorporating advanced feature extraction techniques that improved fraud detection by 65% - Led a cross-functional Agile team of 6 developers through 15+ successful sprint cycles, mentoring 3 junior engineers, while maintaining 95% on time delivery rate	
Reliance Retail Pvt Ltd <i>Automation and Reporting Executive</i>	August 2021 – December 2021 Mumbai, IN
- Automated reporting pipelines with Tableau and SAP that reduced daily reporting time from 4 hours to 30 minutes, enabling real time decision making across 200+ retail locations - Devised software driven automation for picking and scheduling using python to analyze and process data, resulting in 30% reduced warehouse labor costs and 15% increased quarterly profits - Implemented Data analytics dashboards that processes daily transactions, identifying key performance trends that led to the successful reallocation of inventory to new locations to increase delivery efficiency	

Recent Projects

<u>Super Resolution</u>	April 2025
<i>Deep Learning Image Super-Resolution Network PyTorch, Computer Vision, CNNs</i> Enhanced deep residual network with mean shift normalization layers, multi-scale feature extraction, and global residual connections for enhanced gradient flow with modular architecture featuring configurable residual blocks, sub-pixel convolution upsampling, and RGB mean shift preprocessing for improved training stability	
<u>Real-Time Object Detection and Classification System</u>	February 2025
<i>Computer Vision & Machine Learning Pipeline C++, OpenCV</i> Developed real-time object detection system using OpenCV C++ with live camera feed processing and interactive machine learning training. Implemented complete computer vision pipeline featuring adaptive thresholding, morphological operations, and connected component analysis for robust object segmentation	
<u>Augmented Reality</u>	March 2025
<i>Computer Vision & 3D Graphics C++, OpenCV, Camera Calibration</i> Developed real-time augmented reality system using checkerboard pattern detection for accurate 6DOF camera pose estimation and 3D virtual object projection with OpenGL	
<u>Guidance Navigation and Control</u>	April 2025
<i>Aerospace Control System & MATLAB Simulation MATLAB, Control Theory, State-Space Modeling</i> Designed and implemented Linear Quadratic Regulator (LQR) controller for missile longitudinal dynamics with integral augmentation for zero steady-state error tracking. Developed state-space representation of missile dynamics with controllability and observability analysis using MATLAB Control Systems Toolbox	
<u>RRT* Path Planning with Obstacle Avoidance</u>	December 2024
<i>Autonomous Robot Navigation System Python, SLAM, numpy, matplotlib, PIL</i> Revised sampling-based motion planning with real-time obstacle detection and replanning capabilities. System features collision detection using Bresenham line algorithm, near-neighbor search optimization, and cost-based path rewiring for convergence to optimal solutions, achieving robust navigation with a configurable search radius	

Education

Northeastern University <i>Master of Science in Robotics GPA: 3.85/4</i> Mobile Robotics, Computer Vision, Robot Sensing and Navigation, Controls systems	September 2024 – December 2026 Boston, MA
Mumbai University <i>Bachelor's Degree in Mechanical Engineering GPA: 3.5/4</i> Mechatronics, Control System, Kinematics and Dynamics, Machine Learning Achievements Runner Up for Robotics Maze Competition (IIT Bombay)	June 2017 – August 2021 Mumbai, India